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**Task 1: AI-Assisted Personal Portfolio Website**
**Scenario**
**A student wants to showcase academic projects, technical skills, and contact information through a personal portfolio website. AI tools are used to speed up development while maintaining quality and customization.**
**Tasks**
**1. Use GitHub Copilot to generate a semantic HTML structure with sections:**
**o Home**
**o About**
**o Projects**
**o Contact**
**2. Ask Copilot to suggest responsive CSS using Flexbox or Grid.**
**3. Customize AI-generated CSS to add:**
**o Hover effects on project cards**
**o Smooth scrolling navigation**
**Expected Outcome**
**• Clean and well-structured portfolio website**
**• Responsive layout for desktop and mobile screens**
**• Interactive project cards**

## PROMPT:

Create a **personal portfolio website** using **semantic HTML** with sections: Home, About, Projects, and Contact.
Make the layout **responsive using CSS Flexbox or Grid** for both desktop and mobile screens.
Add **project cards with hover effects** and implement **smooth scrolling navigation**.

## CODE:

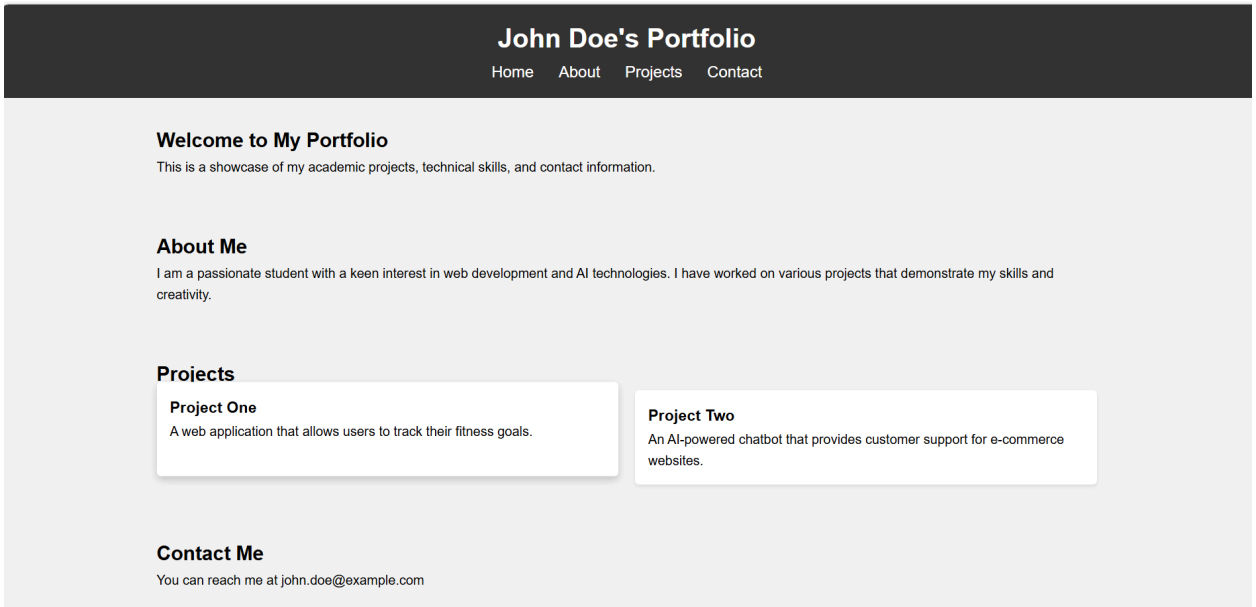
```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Personal Portfolio</title>
    <style>
        /* Basic Reset */
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        body {
            font-family: Arial, sans-serif;
            line-height: 1.6;
            background-color: #f4f4f4;
        }

        header {
            background-color: #333;
            color: #fff;
            padding: 1rem 0;
            text-align: center;
        }

        nav ul {
            list-style: none;
```

```
        display: flex;
        justify-content: center;
    }

    nav ul li {
        margin: 0 15px;
    }

    nav ul li a {
        color: #fff;
        text-decoration: none;
        font-size: 1.2rem;
    }

    section {
        padding: 2rem;
        max-width: 1200px;
        margin: auto;
    }

    .projects {
        display: grid;
        grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
        gap: 20px;
    }

    .project-card {
        background-color: #fff;
        padding: 1rem;
        border-radius: 5px;
        box-shadow: 0 2px 5px rgba(0,0,0,0.1);
        transition: transform 0.3s ease, box-shadow 0.3s ease;
    }

    .project-card:hover {
        transform: translateY(-10px);
        box-shadow: 0 4px 10px rgba(0,0,0,0.2);
    }

    footer {
```

```
            background-color: #333;
            color: #fff;
            text-align: center;
            padding: 1rem 0;
        }
    </style>
</head>
<body>
    <header>
        <h1>John Doe's Portfolio</h1>
        <nav>
            <ul>
                <li><a href="#home">Home</a></li>
                <li><a href="#about">About</a></li>
                <li><a href="#projects">Projects</a></li>
                <li><a href="#contact">Contact</a></li>
            </ul>
        </nav>
    </header>

    <section id="home">
        <h2>Welcome to My Portfolio</h2>
        <p>This is a showcase of my academic projects, technical skills,
and contact information.</p>
    </section>

    <section id="about">
        <h2>About Me</h2>
        <p>I am a passionate student with a keen interest in web
development and AI technologies. I have worked on various projects that
demonstrate my skills and creativity.</p>
    </section>

    <section id="projects">
        <h2>Projects</h2>
        <div class="projects">
            <div class="project-card">
                <h3>Project One</h3>
                <p>A web application that allows users to track their
fitness goals.</p>
```

```html
            </div>
            <div class="project-card">
                <h3>Project Two</h3>
                <p>An AI-powered chatbot that provides customer support
for e-commerce websites.</p>
            </div>
            <!-- Add more project cards as needed -->
        </div>
    </section>

    <section id="contact">
        <h2>Contact Me</h2>
        <p>You can reach me at john.doe@example.com</p>
    </section>

    <footer>
        <p>&copy; 2023 John Doe. All rights reserved.</p>
    </footer>
</body>
</html>
```

# OUTPUT:



## Justification:

1. **Use of Semantic HTML**
   The code uses structured sections such as **header, nav, section, and footer**, which improves readability, accessibility, and SEO. This makes the website easier to maintain and understand.

2. **Responsive Design**
   The **CSS Grid layout in the projects section** (`grid-template-columns: repeat(auto-fit, minmax(300px, 1fr))`) automatically adjusts the number of project cards based on screen size, ensuring the website works well on both desktop and mobile devices.

3. **Interactive User Experience**
   The **hover effect on project cards** using `transform` and `box-shadow` creates visual feedback for users, making the interface more interactive and engaging.

4. **Easy Navigation**
   The navigation bar links (`#home`, `#about`, `#projects`, `#contact`) allow users to quickly move between sections of the webpage, improving usability.

5. **Clean and Maintainable Styling**
   The CSS includes a **basic reset and organized styling**, which ensures consistent layout across browsers and keeps the design simple and professional.

## Task 2: AI-Generated Restaurant Landing Page

## Scenario

## A local restaurant requires a simple yet attractive landing page to display its

## offerings. The developer uses AI assistance to rapidly generate UI components.

## Tasks

## 1. Use Copilot to create a navigation bar with links:

**o Home**

**o Menu**

**o Gallery**

**o Contact**

**2. Generate a menu section styled with CSS cards.**

**3. Implement a JavaScript-based image slider for the gallery using Copilot**

**suggestions.**

**4. Customize animations or transitions for better user experience.**

**Expected Outcome**

**• Fully functional restaurant landing page**

**• Responsive navigation bar**

**• Working image slider with smooth transitions**

**PROMPT:**

Create a restaurant landing page with a navigation bar (Home, Menu, Gallery, Contact) using HTML, CSS, and JavaScript.

Add a menu section with CSS card design and make the layout responsive.

Implement a JavaScript image slider for the gallery with smooth animations and transitions.

## CODE:

```html
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Restaurant Landing Page</title>

    <style>

        /* Basic Reset */

        * {

            margin: 0;

            padding: 0;

            box-sizing: border-box;

        }


        body {

            font-family: 'Arial', sans-serif;

            background-color: #fff8f0;

        }


        header {

            background-color: #d35400;

            color: #fff;
```

```
        padding: 1rem 0;
        text-align: center;
    }

    nav ul {
        list-style: none;
        display: flex;
        justify-content: center;
    }

    nav ul li {
        margin: 0 15px;
    }

    nav ul li a {
        color: #fff;
        text-decoration: none;
        font-size: 1.2rem;
    }

    section {
        padding: 2rem;
        max-width: 1200px;
```

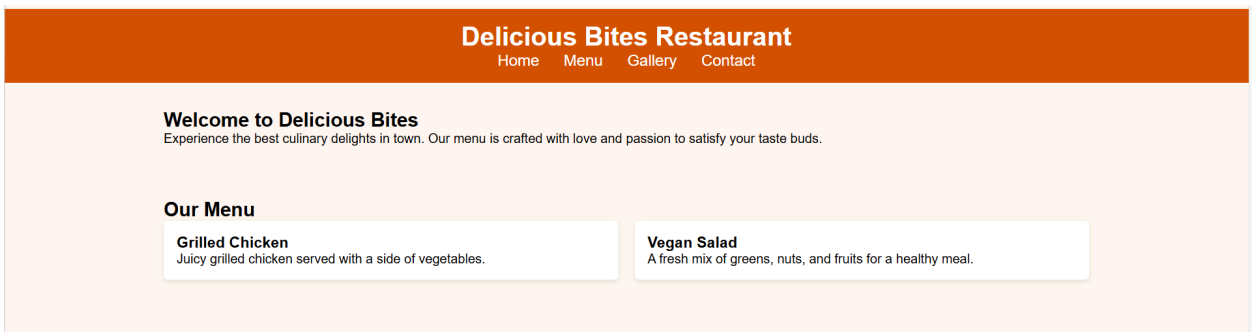
```
        margin: auto;
    }

    .menu-cards {
        display: grid;
        grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
        gap: 20px;
    }

    .menu-card {
        background-color: #fff;
        padding: 1rem;
        border-radius: 5px;
        box-shadow: 0 2px 5px rgba(0,0,0,0.1);
    }

    .slider {
        position: relative;
        width: 100%;
        max-width: 800px;
        margin: auto;
        overflow: hidden;
    }
```

```
        .slides {
            display: flex;
            transition: transform 0.5s ease-in-out;
        }

        .slide {
            min-width: 100%;
        }

        footer {
            background-color: #d35400;
            color: #fff;
            text-align: center;
            padding: 1rem 0;
        }
    </style>
</head>
<body>
    <header>
        <h1>Delicious Bites Restaurant</h1>
        <nav>
            <ul>
```

```
                <li><a href="#home">Home</a></li>

                <li><a href="#menu">Menu</a></li>

                <li><a href="#gallery">Gallery</a></li>

                <li><a href="#contact">Contact</a></li>

            </ul>

        </nav>

    </header>



    <section id="home">

        <h2>Welcome to Delicious Bites</h2>

        <p>Experience the best culinary delights in town. Our menu is
crafted with love and passion to satisfy your taste buds.</p>

    </section>



    <section id="menu">

        <h2>Our Menu</h2>

        <div class="menu-cards">

            <div class="menu-card">

                <h3>Grilled Chicken</h3>

                <p>Juicy grilled chicken served with a side of
vegetables.</p>

            </div>

            <div class="menu-card">

                <h3>Vegan Salad</h3>
```

```
                <p>A fresh mix of greens, nuts, and fruits for a healthy
meal.</p>

            </div>

            <!-- Add more menu cards as needed -->

        </div>

    </section>



</html>
```

## OUTPUT:




## JUSTIFICATION:

The restaurant landing page is designed to provide a simple and attractive interface for users. The responsive navigation bar with links such as Home, Menu, Gallery, and Contact allows users to easily move between different sections of the website. The menu section is displayed using CSS cards, which organize food items in a clear and visually appealing way. The JavaScript-based image slider in the gallery helps showcase images of food and the restaurant environment dynamically. Additionally, smooth animations and transitions improve the user experience and make the webpage look more interactive and professional.

**Task 3: AI-Powered Event Registration Form**

**Scenario**

**SR University is hosting a technical fest and requires an online registration form**

**with real-time validation to ensure accurate user input.**

**Tasks**

**1. Ask Copilot to generate an HTML form with the following fields:**

**o Name**

**o Email**

**o Phone Number**

**o Department**

**o Event Selection**

**2. Use AI assistance to style the form using CSS for readability and**

**accessibility.**

**3. Implement JavaScript validation using Copilot:**

**o Email format validation**

**o Phone number length validation**

**o Required field checks**

**Expected Outcome**

**• User-friendly and visually appealing form**

**• Real-time validation messages**

**• Prevention of invalid submissions**

**PROMPT:**

Create an event registration form using HTML, CSS, and JavaScript with fields: Name, Email, Phone Number, Department, and Event Selection.

Style the form with CSS for good readability and accessibility.

Add JavaScript validation for email format, phone number length, and required fields with real-time error messages.

**CODE:**

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>SR University - Event Registration Form</title>

    <style>

        /* ===== Basic Reset & Global Styles ===== */

        * {

            margin: 0;

            padding: 0;

            box-sizing: border-box;

        }



        html {
```

```
    scroll-behavior: smooth;
}

body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    min-height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    padding: 2rem;
}

/* ===== Container ===== */
.form-container {
    background-color: #fff;
    padding: 3rem;
    border-radius: 12px;
    box-shadow: 0 10px 30px rgba(0, 0, 0, 0.3);
    width: 100%;
    max-width: 500px;
    animation: slideIn 0.5s ease-out;
}
```

```
@keyframes slideIn {
    from {
        opacity: 0;
        transform: translateY(-20px);
    }
    to {
        opacity: 1;
        transform: translateY(0);
    }
}

/* ===== Form Header ===== */
.form-container h1 {
    color: #667eea;
    text-align: center;
    margin-bottom: 1rem;
    font-size: 2rem;
    font-weight: 700;
}

.form-container > p {
    text-align: center;
```

```
        color: #666;
        margin-bottom: 2rem;
        font-size: 0.95rem;
        line-height: 1.5;
    }

    /* ===== Form Structure ===== */
    .registration-form {
        display: flex;
        flex-direction: column;
    }

    .form-group {
        margin-bottom: 1.5rem;
        position: relative;
    }

    .form-group label {
        display: block;
        margin-bottom: 0.6rem;
        color: #333;
        font-weight: 600;
        font-size: 0.95rem;
```

```
}

.required {
    color: #e74c3c;
    margin-left: 0.3rem;
}

/* ===== Input & Select Styling ===== */
.form-group input,
.form-group select {
    width: 100%;
    padding: 0.85rem;
    border: 2px solid #ddd;
    border-radius: 6px;
    font-family: inherit;
    font-size: 1rem;
    transition: all 0.3s ease;
    background-color: #f9f9f9;
}

.form-group input:focus,
.form-group select:focus {
    outline: none;
```

```
        border-color: #667eea;
        background-color: #fff;
        box-shadow: 0 0 8px rgba(102, 126, 234, 0.2);
    }

    .form-group input:disabled {
        background-color: #e8e8e8;
        cursor: not-allowed;
    }

    /* ===== Validation States ===== */
    .form-group input.valid,
    .form-group select.valid {
        border-color: #27ae60;
        background-color: #f0fff4;
    }

    .form-group input.invalid,
    .form-group select.invalid {
        border-color: #e74c3c;
        background-color: #fff5f5;
    }
```

```
/* ===== Validation Messages ===== */

.error-message {

    color: #e74c3c;

    font-size: 0.85rem;

    margin-top: 0.4rem;

    display: none;

    animation: fadeIn 0.3s ease-out;

}



.success-message {

    color: #27ae60;

    font-size: 0.85rem;

    margin-top: 0.4rem;

    display: none;

    animation: fadeIn 0.3s ease-out;

}



.error-message.show,

.success-message.show {

    display: block;

}



@keyframes fadeIn {
```

```
        from {
            opacity: 0;
            transform: translateY(-5px);
        }
        to {
            opacity: 1;
            transform: translateY(0);
        }
    }

    /* ===== Icon for validation ===== */
    .validation-icon {
        position: absolute;
        right: 12px;
        top: 40px;
        font-size: 1.2rem;
        display: none;
    }

    .validation-icon.show {
        display: block;
    }
```

```
/* ===== Submit Button ===== */

.submit-btn {

    padding: 1rem;

    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);

    color: #fff;

    border: none;

    border-radius: 6px;

    font-size: 1rem;

    font-weight: 600;

    cursor: pointer;

    transition: all 0.3s ease;

    margin-top: 1rem;

    box-shadow: 0 4px 12px rgba(102, 126, 234, 0.3);

}


.submit-btn:hover:not(:disabled) {

    transform: translateY(-2px);

    box-shadow: 0 6px 16px rgba(102, 126, 234, 0.4);

}


.submit-btn:active:not(:disabled) {

    transform: translateY(0);

}
```

```
.submit-btn:disabled {
    opacity: 0.6;
    cursor: not-allowed;
}

/* ===== Success Message ===== */
.success-box {
    background-color: #f0fff4;
    border: 2px solid #27ae60;
    border-radius: 6px;
    padding: 1rem;
    margin-bottom: 1.5rem;
    display: none;
    text-align: center;
    animation: slideIn 0.5s ease-out;
}

.success-box.show {
    display: block;
}

.success-box h3 {
```

```
        color: #27ae60;
        margin-bottom: 0.5rem;
    }

    .success-box p {
        color: #27ae60;
        margin: 0;
    }

    /* ===== Responsive Design ===== */
    @media (max-width: 480px) {
        .form-container {
            padding: 2rem 1.5rem;
        }

        .form-container h1 {
            font-size: 1.6rem;
        }

        .submit-btn {
            padding: 0.9rem;
            font-size: 0.95rem;
        }
```

```
        }
    </style>
</head>
<body>
    <div class="form-container">
        <h1>🎓 Event Registration</h1>
        <p>SR University Technical Fest - Register for exciting events</p>


        <div class="success-box" id="successBox">
            <h3>✓ Registration Successful!</h3>
            <p>Thank you for registering. We look forward to seeing you at
the event!</p>
        </div>


        <form class="registration-form" id="eventForm" novalidate>
            <!-- Name Field -->
            <div class="form-group">
                <label for="name">
                    Full Name
                    <span class="required">*</span>
                </label>
                <input
                    type="text"
                    id="name"
```

```
                    name="name"
                    placeholder="Enter your full name"
                    required
                >
                <span class="validation-icon" id="nameIcon"></span>
                <div class="error-message" id="nameError"></div>
                <div class="success-message" id="nameSuccess">✓ Name is
valid</div>
            </div>


            <!-- Email Field -->
            <div class="form-group">
                <label for="email">
                    Email Address
                    <span class="required">*</span>
                </label>
                <input
                    type="email"
                    id="email"
                    name="email"
                    placeholder="Enter your email (e.g.,
user@example.com)"
                    required
                >
```

```
                <span class="validation-icon" id="emailIcon"></span>

                <div class="error-message" id="emailError"></div>

                <div class="success-message" id="emailSuccess">✓ Email
format is valid</div>

            </div>



            <!-- Phone Number Field -->

            <div class="form-group">

                <label for="phone">

                    Phone Number

                    <span class="required">*</span>

                </label>

                <input

                    type="tel"

                    id="phone"

                    name="phone"

                    placeholder="Enter 10-digit phone number"

                    maxlength="10"

                    required

                >

                <span class="validation-icon" id="phoneIcon"></span>

                <div class="error-message" id="phoneError"></div>

                <div class="success-message" id="phoneSuccess">✓ Phone
number is valid</div>
```

```
            </div>


            <!-- Department Field -->
            <div class="form-group">
                <label for="department">
                    Department
                    <span class="required">*</span>
                </label>
                <select id="department" name="department" required>
                    <option value="">-- Select Your Department --</option>
                    <option value="CSE">Computer Science Engineering
(CSE)</option>
                    <option value="ECE">Electronics and Communication
(ECE)</option>
                    <option value="ME">Mechanical Engineering
(ME)</option>
                    <option value="CE">Civil Engineering (CE)</option>
                    <option value="EEE">Electrical and Electronics
(EEE)</option>
                </select>
                <span class="validation-icon" id="departmentIcon"></span>
                <div class="error-message" id="departmentError"></div>
                <div class="success-message" id="departmentSuccess">✓
Department selected</div>
            </div>
```

```
            <!-- Event Selection Field -->
            <div class="form-group">
                <label for="event">
                    Event Selection
                    <span class="required">*</span>
                </label>
                <select id="event" name="event" required>
                    <option value="">-- Select an Event --</option>
                    <option value="coding">Coding Competition</option>
                    <option value="robotics">Robotics Workshop</option>
                    <option value="webdev">Web Development
Challenge</option>
                    <option value="iot">IoT Project Showcase</option>
                    <option value="ai">AI/ML Workshop</option>
                </select>
                <span class="validation-icon" id="eventIcon"></span>
                <div class="error-message" id="eventError"></div>
                <div class="success-message" id="eventSuccess">✓ Event
selected</div>
            </div>


            <!-- Submit Button -->
```

```
            <button type="submit" class="submit-btn"
id="submitBtn">Register Now</button>

        </form>

    </div>



    <script>

        // Get form elements

        const form = document.getElementById('eventForm');

        const nameInput = document.getElementById('name');

        const emailInput = document.getElementById('email');

        const phoneInput = document.getElementById('phone');

        const departmentSelect = document.getElementById('department');

        const eventSelect = document.getElementById('event');

        const successBox = document.getElementById('successBox');

        const submitBtn = document.getElementById('submitBtn');


        // Regular expressions for validation

        const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

        const phoneRegex = /^[0-9]{10}$/;

        const nameRegex = /^[a-zA-Z\s]{3,}$/;


        // Validation functions

        function validateName() {

            const name = nameInput.value.trim();
```

```
            clearMessages('name');


            if (name === '') {
                showError('name', 'Name is required');
                return false;
            } else if (name.length < 3) {
                showError('name', 'Name must be at least 3 characters');
                return false;
            } else if (!nameRegex.test(name)) {
                showError('name', 'Name can only contain letters and
spaces');
                return false;
            } else {
                showSuccess('name');
                return true;
            }
        }


        function validateEmail() {
            const email = emailInput.value.trim();
            clearMessages('email');


            if (email === '') {
                showError('email', 'Email is required');
```

```
                return false;
            } else if (!emailRegex.test(email)) {
                showError('email', 'Please enter a valid email address');
                return false;
            } else {
                showSuccess('email');
                return true;
            }
        }

        function validatePhone() {
            const phone = phoneInput.value.trim();
            clearMessages('phone');

            if (phone === '') {
                showError('phone', 'Phone number is required');
                return false;
            } else if (!phoneRegex.test(phone)) {
                showError('phone', 'Phone number must be exactly 10
digits');
                return false;
            } else {
                showSuccess('phone');
                return true;
```

```
        }
    }

    function validateDepartment() {
        const department = departmentSelect.value;
        clearMessages('department');

        if (department === '') {
            showError('department', 'Please select a department');
            return false;
        } else {
            showSuccess('department');
            return true;
        }
    }

    function validateEvent() {
        const event = eventSelect.value;
        clearMessages('event');

        if (event === '') {
            showError('event', 'Please select an event');
            return false;
```

```
            } else {
                showSuccess('event');
                return true;
            }
        }

        // Helper functions
        function showError(fieldType, message) {
            const input = document.getElementById(fieldType);
            const errorElement = document.getElementById(fieldType +
'Error');
            const icon = document.getElementById(fieldType + 'Icon');

            input.classList.add('invalid');
            input.classList.remove('valid');
            errorElement.textContent = '✗ ' + message;
            errorElement.classList.add('show');
            icon.textContent = '❌';
            icon.classList.add('show');
        }

        function showSuccess(fieldType) {
            const input = document.getElementById(fieldType);
```

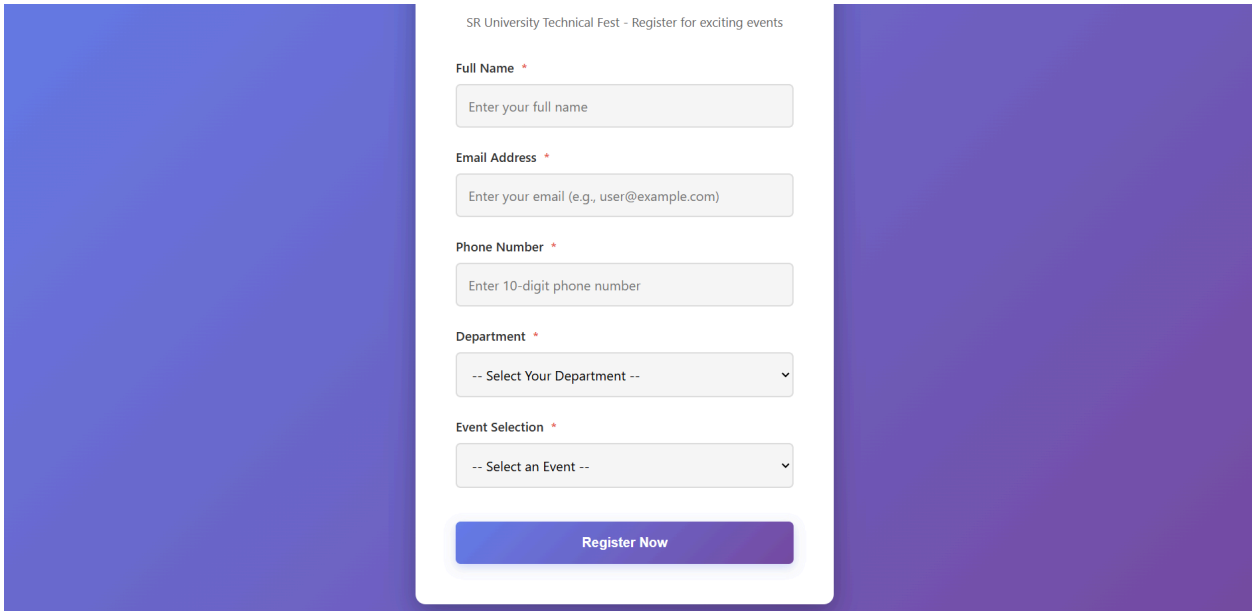
```
            const successElement = document.getElementById(fieldType +
'Success');

            const icon = document.getElementById(fieldType + 'Icon');



            input.classList.add('valid');

            input.classList.remove('invalid');

            successElement.classList.add('show');

            icon.textContent = '✓';

            icon.classList.add('show');

        }



        function clearMessages(fieldType) {

            const errorElement = document.getElementById(fieldType +
'Error');

            const successElement = document.getElementById(fieldType +
'Success');

            const icon = document.getElementById(fieldType + 'Icon');



            errorElement.classList.remove('show');

            successElement.classList.remove('show');

            icon.classList.remove('show');

        }



        // Real-time validation on input
```

```
nameInput.addEventListener('blur', validateName);
nameInput.addEventListener('input', validateName);

emailInput.addEventListener('blur', validateEmail);
emailInput.addEventListener('input', validateEmail);

phoneInput.addEventListener('blur', validatePhone);
phoneInput.addEventListener('input', validatePhone);

departmentSelect.addEventListener('change', validateDepartment);
departmentSelect.addEventListener('blur', validateDepartment);

eventSelect.addEventListener('change', validateEvent);
eventSelect.addEventListener('blur', validateEvent);

// Form submission
form.addEventListener('submit', function(e) {
    e.preventDefault();

    // Validate all fields
    const isNameValid = validateName();
    const isEmailValid = validateEmail();
    const isPhoneValid = validatePhone();
```

```
            const isDepartmentValid = validateDepartment();

            const isEventValid = validateEvent();



            // If all validations pass, show success message

            if (isNameValid && isEmailValid && isPhoneValid &&
isDepartmentValid && isEventValid) {

                successBox.classList.add('show');

                form.style.display = 'none';



                // Reset form after 3 seconds

                setTimeout(() => {

                    form.reset();

                    form.style.display = 'flex';

                    successBox.classList.remove('show');



                    // Clear all validation states

                    document.querySelectorAll('.form-group input,
.form-group select').forEach(field => {

                        field.classList.remove('valid', 'invalid');

                    });

                    document.querySelectorAll('.error-message,
.success-message, .validation-icon').forEach(el => {

                        el.classList.remove('show');

                    });
```

```
                }, 3000);

            }

        });

    </script>

</body>

</html>
```

**OUTPUT:**




## Justification

The event registration form ensures that participants provide correct and complete information before submission. HTML is used to structure the form fields, while CSS improves readability and makes the form visually appealing and user-friendly. JavaScript validation

checks the email format, phone number length, and required fields in real time, preventing invalid data from being submitted and ensuring accurate registration details for the event.

**Task 4: AI-Assisted E-Commerce Product Page**

**Scenario**

**A startup is developing a basic e-commerce product page to showcase products**

**and manage simple cart functionality.**

**Tasks**

**1. Use Copilot to generate a grid-based product catalog using HTML and**

**CSS.**

**2. Implement an “Add to Cart” feature using JavaScript.**

**3. Modify AI-generated code to include:**

**o Cart item counter at the top-right corner**

**o Visual feedback when a product is added**

**Expected Outcome**

**• Functional product catalog**

**• Working cart counter**

**• Interactive user experience**

**PROMPT:**

Create a simple e-commerce product page using HTML, CSS Grid, and JavaScript to display products in a grid layout.

Add an “Add to Cart” button for each product with JavaScript functionality.

Include a cart item counter at the top-right corner and visual feedback when a product is added.

**CODE:**

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>ShopHub - Premium E-Commerce Store</title>

    <style>

        /* ===== Global Styles ===== */

        * {

            margin: 0;

            padding: 0;

            box-sizing: border-box;

        }
```

```
html {
    scroll-behavior: smooth;
}

body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    background-color: #f5f5f5;
    color: #333;
    line-height: 1.6;
}

/* ===== Header ===== */
header {
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    color: white;
    padding: 1rem 0;
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.15);
    position: sticky;
    top: 0;
    z-index: 100;
}

.header-container {
```

```
        max-width: 1200px;
        margin: 0 auto;
        padding: 0 2rem;
        display: flex;
        justify-content: space-between;
        align-items: center;
        flex-wrap: wrap;
        gap: 2rem;
    }

    .logo {
        font-size: 1.8rem;
        font-weight: 700;
        display: flex;
        align-items: center;
        gap: 0.5rem;
    }

    .logo-icon {
        font-size: 2rem;
    }

    /* ===== Navigation ===== */
```

```
nav ul {
    list-style: none;
    display: flex;
    gap: 2rem;
    flex-wrap: wrap;
}

nav ul li a {
    color: white;
    text-decoration: none;
    font-weight: 500;
    transition: opacity 0.3s ease;
}

nav ul li a:hover {
    opacity: 0.8;
}

/* ===== Cart Counter ===== */
.cart-container {
    position: relative;
    cursor: pointer;
}
```

```
.cart-icon {
    font-size: 1.5rem;
    transition: transform 0.3s ease;
}

.cart-icon:hover {
    transform: scale(1.1);
}

.cart-counter {
    position: absolute;
    top: -10px;
    right: -10px;
    background-color: #e74c3c;
    color: white;
    border-radius: 50%;
    width: 28px;
    height: 28px;
    display: flex;
    align-items: center;
    justify-content: center;
    font-weight: 700;
```

```
        font-size: 0.8rem;
        box-shadow: 0 2px 8px rgba(0, 0, 0, 0.2);
        animation: pulse 0.5s ease;
    }

    @keyframes pulse {
        0% { transform: scale(0.8); }
        50% { transform: scale(1.2); }
        100% { transform: scale(1); }
    }

    /* ===== Page Title ===== */
    .page-title {
        max-width: 1200px;
        margin: 3rem auto 2rem;
        padding: 0 2rem;
    }

    .page-title h1 {
        font-size: 2.5rem;
        color: #333;
        margin-bottom: 0.5rem;
    }
```

```
.page-title p {
    color: #666;
    font-size: 1.1rem;
}

/* ===== Filter & Search Section ===== */
.filter-section {
    max-width: 1200px;
    margin: 2rem auto;
    padding: 0 2rem;
    display: flex;
    gap: 1rem;
    flex-wrap: wrap;
    align-items: center;
}

.search-box {
    flex: 1;
    min-width: 250px;
}

.search-box input {
```

```
        width: 100%;
        padding: 0.8rem;
        border: 2px solid #ddd;
        border-radius: 6px;
        font-size: 1rem;
        transition: border-color 0.3s ease;
    }

    .search-box input:focus {
        outline: none;
        border-color: #667eea;
    }

    .sort-box select {
        padding: 0.8rem;
        border: 2px solid #ddd;
        border-radius: 6px;
        font-size: 1rem;
        background-color: white;
        cursor: pointer;
    }

    /* ===== Main Container ===== */
```

```
main {
    max-width: 1200px;
    margin: 0 auto;
    padding: 0 2rem 3rem;
}

/* ===== Product Grid ===== */
.product-catalog {
    display: grid;
    grid-template-columns: repeat(auto-fill, minmax(280px, 1fr));
    gap: 2rem;
    margin-bottom: 2rem;
}

.product-card {
    background-color: white;
    border-radius: 10px;
    overflow: hidden;
    box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);
    transition: all 0.3s ease;
    display: flex;
    flex-direction: column;
    animation: fadeIn 0.5s ease-out;
```

```
}

@keyframes fadeIn {
    from {
        opacity: 0;
        transform: translateY(10px);
    }
    to {
        opacity: 1;
        transform: translateY(0);
    }
}

.product-card:hover {
    transform: translateY(-8px);
    box-shadow: 0 8px 20px rgba(0, 0, 0, 0.15);
}

/* ===== Product Image ===== */
.product-image {
    position: relative;
    width: 100%;
    height: 220px;
```

```
        background: linear-gradient(135deg, #e0e7ff 0%, #f3e8ff 100%);
        display: flex;
        align-items: center;
        justify-content: center;
        overflow: hidden;
    }

    .product-image img {
        width: 100%;
        height: 100%;
        object-fit: cover;
        transition: transform 0.3s ease;
    }

    .product-card:hover .product-image img {
        transform: scale(1.05);
    }

    .product-badge {
        position: absolute;
        top: 12px;
        right: 12px;
        background-color: #e74c3c;
```

```
        color: white;
        padding: 0.4rem 0.8rem;
        border-radius: 20px;
        font-size: 0.8rem;
        font-weight: 600;
    }

    .product-badge.hot {
        background-color: #e67e22;
    }

    .product-badge.new {
        background-color: #27ae60;
    }

    /* ===== Product Details ===== */
    .product-details {
        padding: 1.5rem;
        display: flex;
        flex-direction: column;
        flex-grow: 1;
    }
```

```
.product-category {
    color: #667eea;
    font-size: 0.85rem;
    font-weight: 600;
    text-transform: uppercase;
    letter-spacing: 0.5px;
    margin-bottom: 0.5rem;
}

.product-name {
    font-size: 1.2rem;
    font-weight: 600;
    color: #333;
    margin-bottom: 0.8rem;
    min-height: 2.4rem;
}

.product-description {
    color: #666;
    font-size: 0.9rem;
    margin-bottom: 1rem;
    flex-grow: 1;
}
```

```
/* ===== Rating ===== */
.product-rating {
    display: flex;
    align-items: center;
    gap: 0.5rem;
    margin-bottom: 1rem;
}

.stars {
    color: #f39c12;
    font-size: 0.85rem;
}

.rating-count {
    color: #999;
    font-size: 0.8rem;
}

/* ===== Price & Action ===== */
.product-footer {
    display: flex;
    justify-content: space-between;
```

```
        align-items: center;
        margin-top: 1rem;
        padding-top: 1rem;
        border-top: 1px solid #eee;
    }

    .price-section {
        display: flex;
        flex-direction: column;
        gap: 0.2rem;
    }

    .original-price {
        color: #999;
        text-decoration: line-through;
        font-size: 0.9rem;
    }

    .product-price {
        font-size: 1.5rem;
        font-weight: 700;
        color: #667eea;
    }
```

```
/* ===== Add to Cart Button ===== */
.add-to-cart-btn {
    padding: 0.8rem 1.2rem;
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    color: white;
    border: none;
    border-radius: 6px;
    font-weight: 600;
    cursor: pointer;
    transition: all 0.3s ease;
    font-size: 0.95rem;
    display: flex;
    align-items: center;
    gap: 0.5rem;
    white-space: nowrap;
}

.add-to-cart-btn:hover {
    transform: translateY(-2px);
    box-shadow: 0 4px 12px rgba(102, 126, 234, 0.4);
}
```

```
.add-to-cart-btn:active {
    transform: translateY(0);
}

.add-to-cart-btn.added {
    background: linear-gradient(135deg, #27ae60 0%, #229954 100%);
}

/* ===== Toast Notification ===== */
.toast {
    position: fixed;
    bottom: 20px;
    right: 20px;
    background-color: #27ae60;
    color: white;
    padding: 1rem 1.5rem;
    border-radius: 8px;
    box-shadow: 0 6px 20px rgba(0, 0, 0, 0.3);
    animation: slideInRight 0.4s ease-out;
    z-index: 1000;
    display: flex;
    align-items: center;
    gap: 0.8rem;
```

```
}

@keyframes slideInRight {
    from {
        transform: translateX(400px);
        opacity: 0;
    }
    to {
        transform: translateX(0);
        opacity: 1;
    }
}

@keyframes slideOutRight {
    from {
        transform: translateX(0);
        opacity: 1;
    }
    to {
        transform: translateX(400px);
        opacity: 0;
    }
}
```

```
.toast.remove {
    animation: slideOutRight 0.4s ease-out;
}

.toast-icon {
    font-size: 1.2rem;
}

/* ===== Empty State ===== */
.empty-state {
    grid-column: 1 / -1;
    text-align: center;
    padding: 3rem 2rem;
}

.empty-state-icon {
    font-size: 3rem;
    margin-bottom: 1rem;
}

/* ===== Footer ===== */
footer {
```

```
        background-color: #333;
        color: white;
        text-align: center;
        padding: 2rem;
        margin-top: 3rem;
    }

    /* ===== Responsive Design ===== */
    @media (max-width: 768px) {
        .header-container {
            flex-direction: column;
            gap: 1rem;
        }

        nav ul {
            width: 100%;
            justify-content: center;
            gap: 1rem;
            font-size: 0.9rem;
        }

        .page-title h1 {
            font-size: 1.8rem;
```

```
            }

            .product-catalog {
                grid-template-columns: repeat(auto-fill, minmax(200px,
1fr));
                gap: 1rem;
            }

            .filter-section {
                flex-direction: column;
            }

            .search-box {
                width: 100%;
            }

            .toast {
                bottom: 10px;
                right: 10px;
                left: 10px;
                border-radius: 6px;
            }
        }
```

```
@media (max-width: 480px) {
    .product-catalog {
        grid-template-columns: 1fr;
    }

    .header-container {
        padding: 0 1rem;
    }

    .page-title {
        padding: 0 1rem;
    }

    main {
        padding: 0 1rem 2rem;
    }

    .logo {
        font-size: 1.3rem;
    }

    .page-title h1 {
        font-size: 1.5rem;
```

```
            }

        }

    </style>

</head>

<body>

    <!-- Header -->

    <header>

        <div class="header-container">

            <div class="logo">

                <span class="logo-icon">🛍️</span>

                <span>ShopHub</span>

            </div>

            <nav>

                <ul>

                    <li><a href="#home">Home</a></li>

                    <li><a href="#products">Products</a></li>

                    <li><a href="#deals">Deals</a></li>

                    <li><a href="#about">About</a></li>

                </ul>

            </nav>

            <div class="cart-container" id="cartContainer" title="View
Cart">

                <span class="cart-icon">🛒</span>

                <span class="cart-counter" id="cartCounter">0</span>
```

```html
            </div>
        </div>
    </header>

    <!-- Page Title -->
    <section class="page-title">
        <h1>Featured Products</h1>
        <p>Discover our premium collection of quality products</p>
    </section>

    <!-- Filter Section -->
    <section class="filter-section">
        <div class="search-box">
            <input
                type="text"
                id="searchInput"
                placeholder="🔍 Search products..."
            >
        </div>
        <div class="sort-box">
            <select id="sortSelect">
                <option value="featured">Featured</option>
                <option value="price-low">Price: Low to High</option>
```

```
                <option value="price-high">Price: High to Low</option>
                <option value="newest">Newest</option>
            </select>
        </div>
    </section>

    <!-- Main Content -->
    <main>
        <div class="product-catalog" id="productCatalog">
            <!-- Products will be generated by JavaScript -->
        </div>
    </main>

    <!-- Footer -->
    <footer>
        <p>&copy; 2024 ShopHub. All rights reserved. | Premium E-Commerce
Platform</p>
    </footer>

    <script>
        // Product Data
        const products = [
            {
                id: 1,
```

```
        name: 'Premium Wireless Headphones',
        category: 'Electronics',
        price: 149.99,
        originalPrice: 199.99,
        rating: 4.5,
        reviews: 128,
        badge: 'hot',
        description: 'High-quality sound with noise cancellation'
    },
    {
        id: 2,
        name: 'Smartphone Stand',
        category: 'Accessories',
        price: 24.99,
        originalPrice: 34.99,
        rating: 4.8,
        reviews: 245,
        badge: 'new',
        description: 'Adjustable stand for all devices'
    },
    {
        id: 3,
        name: 'USB-C Cable (3 Pack)',
```

```
            category: 'Cables',
            price: 19.99,
            originalPrice: 29.99,
            rating: 4.6,
            reviews: 512,
            badge: 'sale',
            description: 'Durable and fast charging cables'
        },
        {
            id: 4,
            name: '4K Webcam',
            category: 'Electronics',
            price: 89.99,
            originalPrice: 129.99,
            rating: 4.7,
            reviews: 89,
            badge: 'new',
            description: 'Crystal clear video for streaming'
        },
        {
            id: 5,
            name: 'Mechanical Keyboard RGB',
            category: 'Gaming',
```

```
            price: 129.99,
            originalPrice: 169.99,
            rating: 4.9,
            reviews: 342,
            badge: 'hot',
            description: 'Premium gaming keyboard with RGB lighting'
        },
        {
            id: 6,
            name: 'Wireless Mouse Pro',
            category: 'Accessories',
            price: 59.99,
            originalPrice: 79.99,
            rating: 4.4,
            reviews: 176,
            badge: '',
            description: 'Precision mouse with long battery life'
        },
        {
            id: 7,
            name: 'Monitor Light Bar',
            category: 'Lighting',
            price: 44.99,
```

```
        originalPrice: 64.99,
        rating: 4.6,
        reviews: 94,
        badge: 'sale',
        description: 'Smart monitor light for eye care'
    },
    {
        id: 8,
        name: 'Portable SSD 1TB',
        category: 'Storage',
        price: 99.99,
        originalPrice: 149.99,
        rating: 4.8,
        reviews: 267,
        badge: 'new',
        description: 'Fast and reliable external storage'
    },
    {
        id: 9,
        name: 'Phone Cooling Fan',
        category: 'Cooling',
        price: 34.99,
        originalPrice: 49.99,
```

```
        rating: 4.5,
        reviews: 156,
        badge: 'hot',
        description: 'Keep your phone cool during gaming'
    },
    {
        id: 10,
        name: 'Desk Organizer Set',
        category: 'Office',
        price: 29.99,
        originalPrice: 44.99,
        rating: 4.7,
        reviews: 203,
        badge: '',
        description: 'Complete desk organization solution'
    },
    {
        id: 11,
        name: 'HDMI Cable 2.1 (2m)',
        category: 'Cables',
        price: 14.99,
        originalPrice: 24.99,
        rating: 4.6,
```

```
            reviews: 421,
            badge: 'new',
            description: '8K support with gold-plated connectors'
        },
        {
            id: 12,
            name: 'Phone Tempered Glass',
            category: 'Protection',
            price: 12.99,
            originalPrice: 19.99,
            rating: 4.8,
            reviews: 589,
            badge: 'sale',
            description: 'Premium screen protector set (3-pack)'
        }
    ];

    // Cart state
    let cart = [];
    const cartCounter = document.getElementById('cartCounter');
    const productCatalog = document.getElementById('productCatalog');
    const searchInput = document.getElementById('searchInput');
    const sortSelect = document.getElementById('sortSelect');
```

```
        // Initialize
        function init() {
            loadCart();
            renderProducts(products);
            setupEventListeners();
        }


        // Render products
        function renderProducts(productsToRender) {
            if (productsToRender.length === 0) {
                productCatalog.innerHTML = '<div class="empty-state"><div
class="empty-state-icon">🔍</div><p>No products found</p></div>';
                return;
            }


            productCatalog.innerHTML = productsToRender.map(product => `
                <div class="product-card">
                    <div class="product-image">
                        <div style="font-size: 4rem; width: 100%; height:
100%; display: flex; align-items: center; justify-content: center;
background: linear-gradient(135deg, #667eea 0%, #764ba2 100%); color:
white;">
                            ${getProductEmoji(product.category)}
                        </div>
```

```
                        ${product.badge ? `<span class="product-badge
${product.badge}">${product.badge.toUpperCase()}</span>` : ''}

                    </div>

                    <div class="product-details">

                        <div
class="product-category">${product.category}</div>

                        <h3 class="product-name">${product.name}</h3>

                        <p
class="product-description">${product.description}</p>

                        <div class="product-rating">

                            <span
class="stars">${'⭐'.repeat(Math.floor(product.rating))}</span>

                            <span class="rating-count">${product.rating}
(${product.reviews})</span>

                        </div>

                        <div class="product-footer">

                            <div class="price-section">

                                <span
class="original-price">$${product.originalPrice}</span>

                                <span
class="product-price">$${product.price}</span>

                            </div>

                            <button class="add-to-cart-btn"
data-product-id="${product.id}">

                                <span>🛒</span>

                                <span>Add</span>

                            </button>
```

```
                    </div>
                </div>
            </div>
        `).join('');

        attachCartButtonListeners();
    }

    // Get emoji based on category
    function getProductEmoji(category) {
        const emojiMap = {
            'Electronics': '💻',
            'Accessories': '🎧',
            'Cables': '🔌',
            'Gaming': '🎮',
            'Lighting': '💡',
            'Storage': '💾',
            'Cooling': '❄️',
            'Office': '📎',
            'Protection': '🛡️'
        };
        return emojiMap[category] || '📦';
    }
```

```
        // Attach event listeners to cart buttons
        function attachCartButtonListeners() {
            document.querySelectorAll('.add-to-cart-btn').forEach(button
=> {
                button.addEventListener('click', function() {
                    const productId = parseInt(this.dataset.productId);
                    addToCart(productId);

                    // Visual feedback
                    this.classList.add('added');
                    this.textContent = '✓ Added';

                    setTimeout(() => {
                        this.classList.remove('added');
                        this.innerHTML =
'<span>🛒</span><span>Add</span>';
                    }, 2000);
                });
            });
        }

        // Add to cart
        function addToCart(productId) {
```

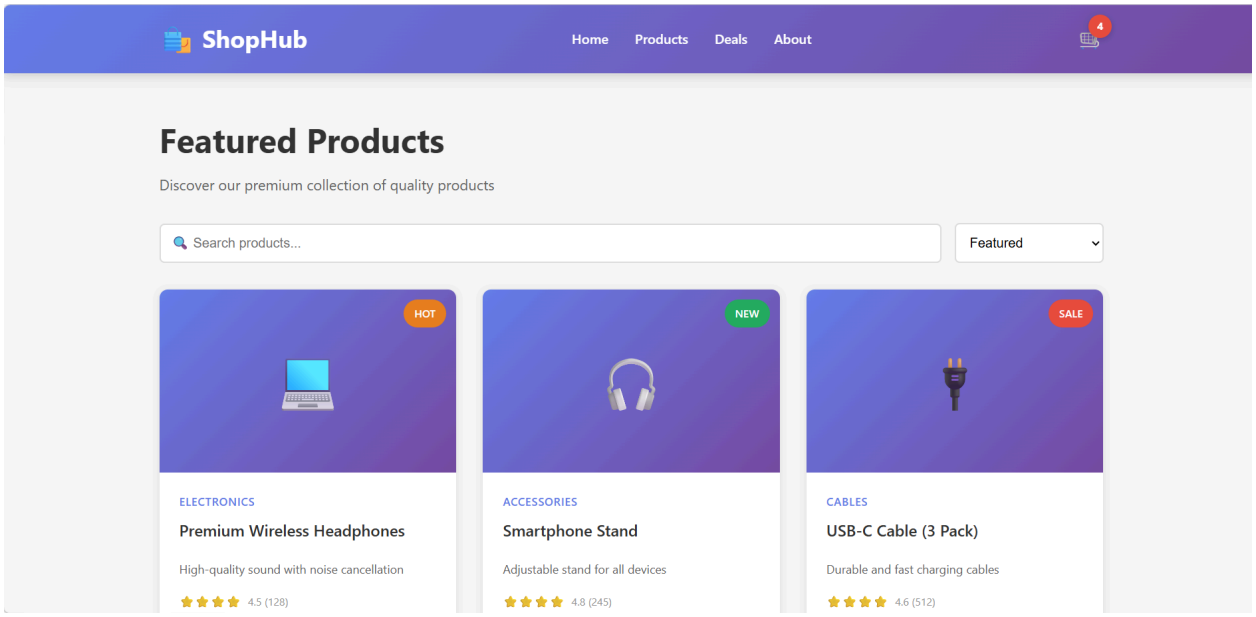
```
            const product = products.find(p => p.id === productId);
            if (!product) return;

            cart.push(product);
            updateCartCounter();
            saveCart();
            showToast(`${product.name} added to cart!`);
        }

        // Update cart counter
        function updateCartCounter() {
            cartCounter.textContent = cart.length;
        }

        // Show toast notification
        function showToast(message) {
            const toast = document.createElement('div');
            toast.className = 'toast';
            toast.innerHTML = `<span
class="toast-icon">✓</span><span>${message}</span>`;
            document.body.appendChild(toast);

            setTimeout(() => {
                toast.classList.add('remove');
```

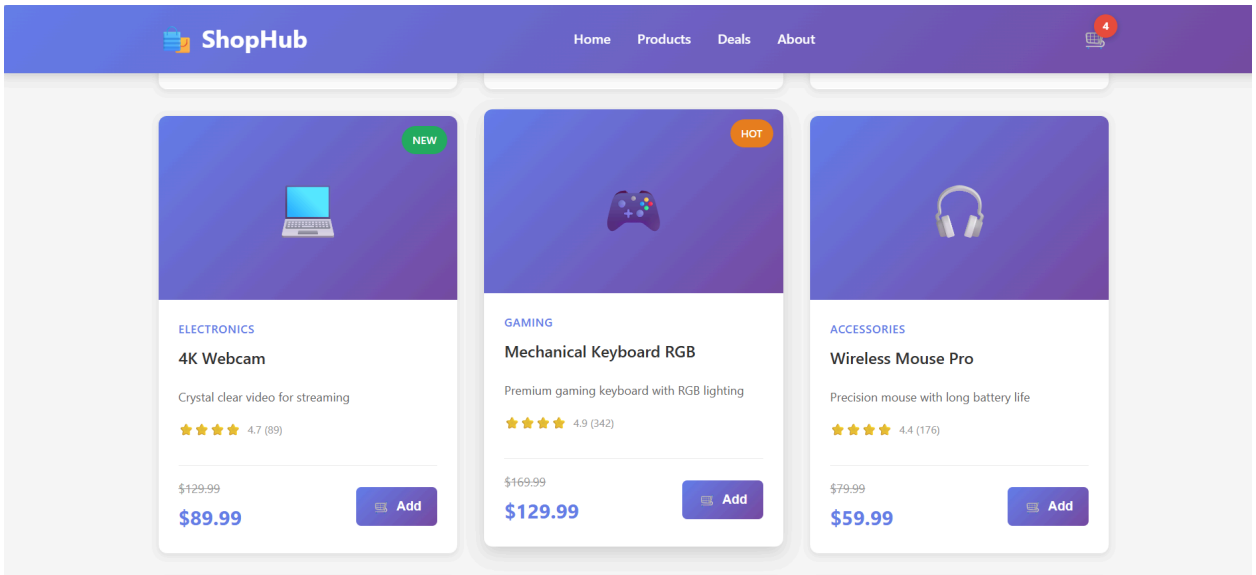
```
            setTimeout(() => toast.remove(), 400);
        }, 2500);
    }

    // Save cart to localStorage
    function saveCart() {
        localStorage.setItem('cart', JSON.stringify(cart));
    }

    // Load cart from localStorage
    function loadCart() {
        const savedCart = localStorage.getItem('cart');
        if (savedCart) {
            cart = JSON.parse(savedCart);
            updateCartCounter();
        }
    }

    // Setup event listeners
    function setupEventListeners() {
        // Search
        searchInput.addEventListener('input', function() {
            const searchTerm = this.value.toLowerCase();
```

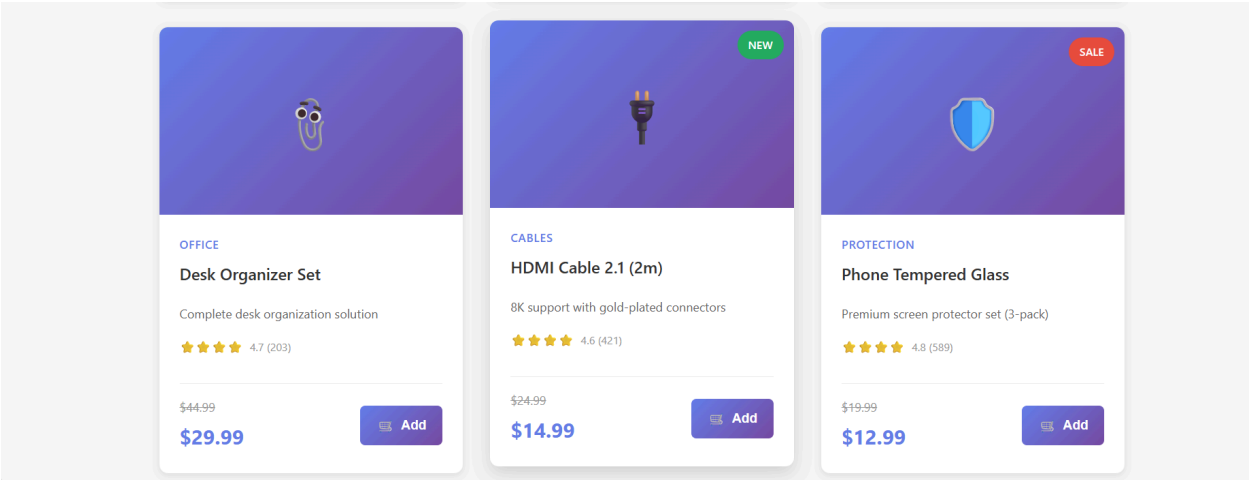
```
        const filtered = products.filter(p =>
            p.name.toLowerCase().includes(searchTerm) ||
            p.category.toLowerCase().includes(searchTerm)
        );
        renderProducts(filtered);
    });

    // Sort
    sortSelect.addEventListener('change', function() {
        let sorted = [...products];

        switch(this.value) {
            case 'price-low':
                sorted.sort((a, b) => a.price - b.price);
                break;
            case 'price-high':
                sorted.sort((a, b) => b.price - a.price);
                break;
            case 'newest':
                sorted.reverse();
                break;
            case 'featured':
            default:
```

```
                        sorted = [...products];

                }



                renderProducts(sorted);

            });



            // Cart click


document.getElementById('cartContainer').addEventListener('click',
function() {

                if (cart.length > 0) {

                    showToast(`Cart contains ${cart.length} item(s)`);

                } else {

                    showToast('Your cart is empty');

                }

            });

        }



        // Start the application

        init();

    </script>

</body>

</html>
```

# OUTPUT:


ShopHub
Home
Products
Deals
About
4
Featured Products
Discover our premium collection of quality products
Search products...
Featured
HOT
NEW
SALE
ELECTRONICS
Premium Wireless Headphones
High-quality sound with noise cancellation
4.5 (128)
ACCESSORIES
Smartphone Stand
Adjustable stand for all devices
4.8 (245)
CABLES
USB-C Cable (3 Pack)
Durable and fast charging cables
4.6 (512)


ShopHub
Home
Products
Deals
About
4
NEW
HOT
ELECTRONICS
4K Webcam
Crystal clear video for streaming
4.7 (89)
$129.99
$89.99
Add
GAMING
Mechanical Keyboard RGB
Premium gaming keyboard with RGB lighting
4.9 (342)
$169.99
$129.99
Add
ACCESSORIES
Wireless Mouse Pro
Precision mouse with long battery life
4.4 (176)
$79.99
$59.99
Add




**Justification:**

The e-commerce product page helps display products in an organized grid layout using HTML and CSS. The JavaScript-based “Add to Cart” feature allows users to select products easily. The cart item counter updates dynamically to show the number of items added, while visual feedback confirms the action, improving the overall user experience and interactivity of the page.

**Task 5: Create a Responsive Web Page Layout**

**Instructions:**

- **Design a basic web page layout with a header, main content area, and footer using HTML and CSS.**
- **Use AI to assist in generating responsive CSS for different screen sizes.**
- **Ensure the layout is clean and visually organized.**

**Expected Output:**

- **A responsive web page with:**
- **Header with navigation links**
- **Main section with placeholder text/images**
- **Footer with copyright or contact info**
- **Layout adapts correctly to desktop, tablet, and mobile screen sizes**

**PROMPT:**

**CODE:**

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Responsive Web Page Layout</title>

    <style>

        /* ===== Global Styles ===== */

        * {

            margin: 0;

            padding: 0;

            box-sizing: border-box;
```

```
}

html {
    scroll-behavior: smooth;
}

body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    line-height: 1.6;
    color: #333;
    background-color: #f8f9fa;
}

/* ===== Header ===== */
header {
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    color: white;
    padding: 1rem 0;
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.15);
    position: sticky;
    top: 0;
    z-index: 100;
}
```

```
.header-container {
    max-width: 1200px;
    margin: 0 auto;
    padding: 0 2rem;
    display: flex;
    justify-content: space-between;
    align-items: center;
    flex-wrap: wrap;
    gap: 1rem;
}

.logo {
    font-size: 1.8rem;
    font-weight: 700;
    display: flex;
    align-items: center;
    gap: 0.5rem;
}

.logo-icon {
    font-size: 2rem;
}
```

```
/* ===== Navigation ===== */
nav {
    display: flex;
    align-items: center;
}

.nav-toggle {
    display: none;
    background: none;
    border: none;
    color: white;
    font-size: 1.5rem;
    cursor: pointer;
    padding: 0.5rem;
}

nav ul {
    list-style: none;
    display: flex;
    gap: 2rem;
    flex-wrap: wrap;
}
```

```
nav ul li a {
    color: white;
    text-decoration: none;
    font-weight: 500;
    padding: 0.5rem 1rem;
    border-radius: 5px;
    transition: all 0.3s ease;
}

nav ul li a:hover {
    background-color: rgba(255, 255, 255, 0.2);
    transform: translateY(-2px);
}

/* ===== Main Content ===== */
main {
    max-width: 1200px;
    margin: 0 auto;
    padding: 2rem;
}

/* ===== Hero Section ===== */
```

```
.hero {
    background: linear-gradient(135deg, #f093fb 0%, #f5576c 100%);
    color: white;
    padding: 4rem 2rem;
    text-align: center;
    border-radius: 10px;
    margin-bottom: 3rem;
    box-shadow: 0 8px 20px rgba(0, 0, 0, 0.1);
}

.hero h1 {
    font-size: 3rem;
    margin-bottom: 1rem;
    font-weight: 700;
}

.hero p {
    font-size: 1.2rem;
    margin-bottom: 2rem;
    max-width: 600px;
    margin-left: auto;
    margin-right: auto;
}
```

```
.cta-button {
    display: inline-block;
    background-color: white;
    color: #f5576c;
    padding: 1rem 2rem;
    text-decoration: none;
    border-radius: 5px;
    font-weight: 600;
    transition: all 0.3s ease;
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.2);
}

.cta-button:hover {
    transform: translateY(-3px);
    box-shadow: 0 6px 16px rgba(0, 0, 0, 0.3);
}

/* ===== Content Sections ===== */
.content-section {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 3rem;
```

```
        margin-bottom: 3rem;
        align-items: center;
    }

    .content-section.reverse {
        grid-template-columns: 1fr 1fr;
    }

    .content-text {
        padding: 2rem;
    }

    .content-text h2 {
        font-size: 2.5rem;
        color: #667eea;
        margin-bottom: 1rem;
        font-weight: 700;
    }

    .content-text p {
        font-size: 1.1rem;
        color: #666;
        margin-bottom: 1.5rem;
```

```
        line-height: 1.8;
    }

    .content-text ul {
        list-style: none;
        padding: 0;
    }

    .content-text ul li {
        padding: 0.5rem 0;
        color: #555;
    }

    .content-text ul li::before {
        content: "✓";
        color: #27ae60;
        font-weight: bold;
        margin-right: 0.5rem;
    }

    .content-image {
        background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
        border-radius: 10px;
```

```
        padding: 2rem;
        text-align: center;
        color: white;
        font-size: 4rem;
        display: flex;
        align-items: center;
        justify-content: center;
        min-height: 300px;
        box-shadow: 0 8px 20px rgba(0, 0, 0, 0.1);
    }

    /* ===== Features Grid ===== */
    .features {
        display: grid;
        grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
        gap: 2rem;
        margin-bottom: 3rem;
    }

    .feature-card {
        background-color: white;
        padding: 2rem;
        border-radius: 10px;
```

```
        box-shadow: 0 4px 12px rgba(0, 0, 0, 0.1);
        text-align: center;
        transition: all 0.3s ease;
    }

    .feature-card:hover {
        transform: translateY(-5px);
        box-shadow: 0 8px 20px rgba(0, 0, 0, 0.15);
    }

    .feature-icon {
        font-size: 3rem;
        margin-bottom: 1rem;
    }

    .feature-card h3 {
        font-size: 1.5rem;
        color: #667eea;
        margin-bottom: 1rem;
        font-weight: 600;
    }

    .feature-card p {
```

```
        color: #666;
        line-height: 1.6;
    }

    /* ===== Stats Section ===== */
    .stats {
        background-color: #667eea;
        color: white;
        padding: 3rem 2rem;
        border-radius: 10px;
        margin-bottom: 3rem;
        text-align: center;
    }

    .stats-grid {
        display: grid;
        grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
        gap: 2rem;
        margin-top: 2rem;
    }

    .stat-item h3 {
        font-size: 2.5rem;
```

```
        font-weight: 700;
        margin-bottom: 0.5rem;
    }

    .stat-item p {
        font-size: 1.1rem;
        opacity: 0.9;
    }

    /* ===== Footer ===== */
    footer {
        background-color: #333;
        color: white;
        padding: 3rem 0 2rem;
        margin-top: 3rem;
    }

    .footer-container {
        max-width: 1200px;
        margin: 0 auto;
        padding: 0 2rem;
        display: grid;
        grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
```

```
        gap: 2rem;
    }

    .footer-section h3 {
        color: #667eea;
        margin-bottom: 1rem;
        font-size: 1.3rem;
    }

    .footer-section p,
    .footer-section ul li {
        color: #ccc;
        margin-bottom: 0.5rem;
        line-height: 1.6;
    }

    .footer-section ul {
        list-style: none;
    }

    .footer-section ul li a {
        color: #ccc;
        text-decoration: none;
```

```
        transition: color 0.3s ease;
    }

    .footer-section ul li a:hover {
        color: #667eea;
    }

    .social-links {
        display: flex;
        gap: 1rem;
        margin-top: 1rem;
    }

    .social-links a {
        display: inline-block;
        width: 40px;
        height: 40px;
        background-color: #667eea;
        color: white;
        text-align: center;
        line-height: 40px;
        border-radius: 50%;
        text-decoration: none;
```

```
        transition: all 0.3s ease;
    }

    .social-links a:hover {
        transform: translateY(-3px);
        box-shadow: 0 4px 12px rgba(102, 126, 234, 0.4);
    }

    .copyright {
        text-align: center;
        margin-top: 2rem;
        padding-top: 2rem;
        border-top: 1px solid #555;
        color: #999;
    }

    /* ===== Responsive Design ===== */

    /* Tablet Styles */
    @media (max-width: 768px) {
        .header-container {
            flex-direction: column;
            gap: 1rem;
```

```
}

.nav-toggle {
    display: block;
    order: 2;
}

nav ul {
    display: none;
    flex-direction: column;
    width: 100%;
    gap: 0.5rem;
    margin-top: 1rem;
}

nav ul.show {
    display: flex;
}

nav ul li a {
    text-align: center;
    padding: 0.8rem;
}
```

```
.hero {
    padding: 3rem 1rem;
}

.hero h1 {
    font-size: 2.2rem;
}

.hero p {
    font-size: 1rem;
}

.content-section {
    grid-template-columns: 1fr;
    gap: 2rem;
}

.content-section.reverse {
    grid-template-columns: 1fr;
}

.content-text {
```

```
                padding: 1rem;

                text-align: center;

            }



            .content-text h2 {

                font-size: 2rem;

            }



            .features {

                grid-template-columns: repeat(auto-fit, minmax(200px,
1fr));

                gap: 1.5rem;

            }



            .stats-grid {

                grid-template-columns: repeat(2, 1fr);

            }



            .footer-container {

                grid-template-columns: 1fr;

                text-align: center;

            }



            .social-links {
```

```
            justify-content: center;
        }
    }

    /* Mobile Styles */
    @media (max-width: 480px) {
        .header-container {
            padding: 0 1rem;
        }

        .logo {
            font-size: 1.5rem;
        }

        main {
            padding: 1rem;
        }

        .hero {
            padding: 2rem 1rem;
            margin-bottom: 2rem;
        }
```

```
.hero h1 {
    font-size: 1.8rem;
}

.hero p {
    font-size: 0.95rem;
}

.content-text {
    padding: 0.5rem;
}

.content-text h2 {
    font-size: 1.6rem;
}

.content-text p {
    font-size: 1rem;
}

.features {
    grid-template-columns: 1fr;
    gap: 1rem;
```

```
}

.feature-card {
    padding: 1.5rem;
}

.feature-icon {
    font-size: 2.5rem;
}

.stats {
    padding: 2rem 1rem;
}

.stats-grid {
    grid-template-columns: 1fr;
}

.stat-item h3 {
    font-size: 2rem;
}

.footer-container {
```

```
            padding: 0 1rem;
        }

        .social-links {
            gap: 0.8rem;
        }

        .social-links a {
            width: 35px;
            height: 35px;
            line-height: 35px;
        }
    }

    /* ===== Animations ===== */
    @keyframes fadeInUp {
        from {
            opacity: 0;
            transform: translateY(30px);
        }
        to {
            opacity: 1;
            transform: translateY(0);
```

```
            }
        }

        .fade-in-up {
            animation: fadeInUp 0.6s ease-out;
        }

        /* ===== Utility Classes ===== */
        .text-center {
            text-align: center;
        }

        .mb-2 {
            margin-bottom: 1rem;
        }

        .mb-3 {
            margin-bottom: 1.5rem;
        }
    </style>
</head>
<body>
    <!-- Header -->
```

```html
<header>
    <div class="header-container">
        <div class="logo">
            <span class="logo-icon">🚀</span>
            <span>TechCorp</span>
        </div>
        <nav>
            <button class="nav-toggle" id="navToggle">☰</button>
            <ul id="navMenu">
                <li><a href="#home">Home</a></li>
                <li><a href="#about">About</a></li>
                <li><a href="#services">Services</a></li>
                <li><a href="#contact">Contact</a></li>
            </ul>
        </nav>
    </div>
</header>


<!-- Main Content -->
<main>
    <!-- Hero Section -->
    <section class="hero" id="home">
        <h1>Welcome to TechCorp</h1>
```

```
            <p>Innovative solutions for modern businesses. We create
cutting-edge technology that drives success and transforms industries.</p>

            <a href="#contact" class="cta-button">Get Started</a>

        </section>



        <!-- About Section -->

        <section class="content-section fade-in-up">

            <div class="content-text">

                <h2>About Our Company</h2>

                <p>TechCorp is a leading technology company specializing
in innovative software solutions, digital transformation, and cutting-edge
web development. With over a decade of experience, we have helped hundreds
of businesses achieve their digital goals.</p>

                <p>Our team of expert developers, designers, and
strategists work together to create solutions that not only meet current
needs but also anticipate future challenges.</p>

                <ul>

                    <li>Expert team of developers and designers</li>

                    <li>Proven track record of successful projects</li>

                    <li>Commitment to quality and innovation</li>

                    <li>24/7 customer support and maintenance</li>

                </ul>

            </div>

            <div class="content-image">

                📊📈

            </div>
```

```
        </section>


        <!-- Services Section -->

        <section class="content-section reverse fade-in-up">

            <div class="content-image">

                💻🔧

            </div>

            <div class="content-text">

                <h2>Our Services</h2>

                <p>We offer comprehensive technology solutions tailored to
meet the unique needs of your business. From web development to mobile
applications, we provide end-to-end services that drive results.</p>

                <p>Our services include custom software development, cloud
solutions, digital marketing, and ongoing support to ensure your business
stays ahead of the competition.</p>

                <ul>

                    <li>Custom web application development</li>

                    <li>Mobile app development (iOS & Android)</li>

                    <li>Cloud infrastructure and migration</li>

                    <li>UI/UX design and prototyping</li>

                    <li>Digital marketing and SEO</li>

                </ul>

            </div>

        </section>
```

```html
        <!-- Features -->

        <section class="features fade-in-up">

            <div class="feature-card">

                <div class="feature-icon">⚡</div>

                <h3>Fast Performance</h3>

                <p>Lightning-fast loading times and optimized performance
ensure your users have the best experience possible.</p>

            </div>

            <div class="feature-card">

                <div class="feature-icon">🔒</div>

                <h3>Secure & Reliable</h3>

                <p>Enterprise-grade security measures protect your data
and ensure 99.9% uptime for your applications.</p>

            </div>

            <div class="feature-card">

                <div class="feature-icon">📱</div>

                <h3>Mobile Responsive</h3>

                <p>All our solutions are fully responsive and work
perfectly across all devices and screen sizes.</p>

            </div>

            <div class="feature-card">

                <div class="feature-icon">🎯</div>

                <h3>Results Driven</h3>

                <p>We focus on delivering measurable results that drive
business growth and customer satisfaction.</p>
```

```
        </div>
    </section>


    <!-- Stats Section -->
    <section class="stats fade-in-up">
        <h2>Our Achievements</h2>
        <div class="stats-grid">
            <div class="stat-item">
                <h3>500+</h3>
                <p>Projects Completed</p>
            </div>
            <div class="stat-item">
                <h3>250+</h3>
                <p>Happy Clients</p>
            </div>
            <div class="stat-item">
                <h3>10+</h3>
                <p>Years Experience</p>
            </div>
            <div class="stat-item">
                <h3>24/7</h3>
                <p>Support Available</p>
            </div>
```

```html
            </div>
        </section>
    </main>


    <!-- Footer -->
    <footer>
        <div class="footer-container">
            <div class="footer-section">
                <h3>TechCorp</h3>
                <p>Innovative technology solutions for modern businesses.
We create digital experiences that drive success.</p>
                <div class="social-links">
                    <a href="#" title="Facebook">📘</a>
                    <a href="#" title="Twitter">🐦</a>
                    <a href="#" title="LinkedIn">💼</a>
                    <a href="#" title="Instagram">📷</a>
                </div>
            </div>
            <div class="footer-section">
                <h3>Quick Links</h3>
                <ul>
                    <li><a href="#home">Home</a></li>
                    <li><a href="#about">About Us</a></li>
                    <li><a href="#services">Services</a></li>
```

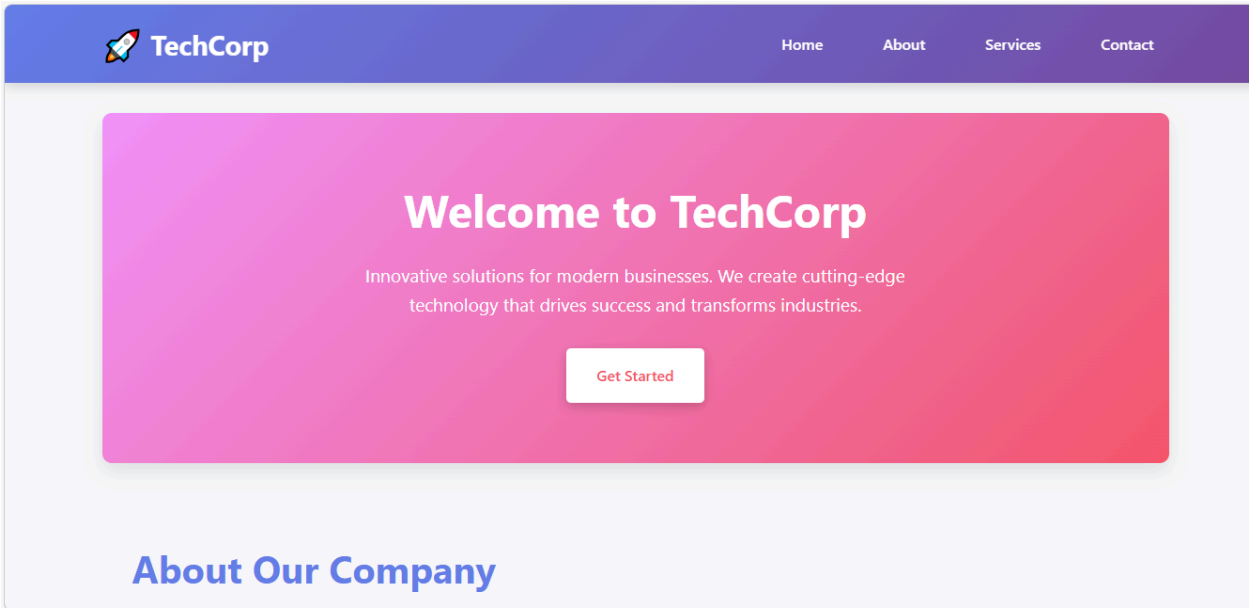
```
                <li><a href="#contact">Contact</a></li>
            </ul>
        </div>
        <div class="footer-section">
            <h3>Services</h3>
            <ul>
                <li><a href="#">Web Development</a></li>
                <li><a href="#">Mobile Apps</a></li>
                <li><a href="#">Cloud Solutions</a></li>
                <li><a href="#">Digital Marketing</a></li>
            </ul>
        </div>
        <div class="footer-section">
            <h3>Contact Info</h3>
            <p>📧 info@techcorp.com</p>
            <p>📞 +1 (555) 123-4567</p>
            <p>📍 123 Tech Street, Silicon Valley, CA 94000</p>
        </div>
    </div>
    <div class="copyright">
        <p>&copy; 2024 TechCorp. All rights reserved. | Designed with ❤️ using AI assistance</p>
    </div>
</footer>
```

```
<script>
    // Mobile navigation toggle
    const navToggle = document.getElementById('navToggle');
    const navMenu = document.getElementById('navMenu');

    navToggle.addEventListener('click', function() {
        navMenu.classList.toggle('show');
    });

    // Smooth scrolling for navigation links
    document.querySelectorAll('nav ul li a').forEach(link => {
        link.addEventListener('click', function(e) {
            e.preventDefault();
            const targetId = this.getAttribute('href');
            const targetSection = document.querySelector(targetId);

            if (targetSection) {
                targetSection.scrollIntoView({
                    behavior: 'smooth',
                    block: 'start'
                });
            }
```

```
        // Close mobile menu after clicking
        navMenu.classList.remove('show');
    });
});

// Add fade-in animation on scroll
const observerOptions = {
    threshold: 0.1,
    rootMargin: '0px 0px -50px 0px'
};

const observer = new IntersectionObserver(function(entries) {
    entries.forEach(entry => {
        if (entry.isIntersecting) {
            entry.target.classList.add('fade-in-up');
        }
    });
}, observerOptions);

// Observe all sections with fade-in-up class
document.querySelectorAll('.fade-in-up').forEach(section => {
    observer.observe(section);
```

```
        });

    </script>

</body>

</html>
```

**OUTPUT:**




## Justification

This task demonstrates how a basic web page layout can be structured using HTML and styled with CSS. The header provides navigation links for easy access to different sections, while the main content area displays information or images. The footer contains copyright or contact details. Responsive CSS ensures that the layout adjusts properly to different screen sizes such as desktop, tablet, and mobile, providing a clean and user-friendly design.